

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

FOOTBALL PLAYER PERFORMANCE DATA ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework of Data Visualization and Analytics (FDVA), focuses on Football Player Performance Data Analysis and Visualization using Microsoft Power BI. The project analyses football player statistics to understand player positions, goals, assists, club performance, and goalkeeper statistics. The dataset is used to create meaningful visual representations that make football performance data easier to understand.Power BI was used to develop an interactive dashboard containing different visualizations, including pie charts, funnel charts, scatter plots, line charts, treemaps, and bar charts. These visualizations help compare player positions, analyse the relationship between goals and assists, examine goals by club, and understand squad-wise statistics.The dashboard presents football statistics in a clear and organized manner, allowing users to explore player and team performance through visual comparisons. The project demonstrates how data visualization can transform raw football statistics into useful insights and support better understanding of player performance and club statistics.

1. Data Preprocessing

The dataset (sportsAnalysis.csv) has 500 rows and 28columns. It was cleaned in Python using Pandas, as shown below.

1.1 Loading and inspecting the data

```
import pandas as pd
df = pd.read_csv("football_data.csv")
print(df.shape)
print(df.info())
print(df.head())
```

Price, discount, EMI months, waiting days and sale date were stored as text and had to be converted.

1.2 Checking Missing Values and Duplicates

```
print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
```

1.3 Checking Numerical Statistics

```
print(df.describe()
```

1.4 Selecting Relevant Columns

```
 selected_columns = [
    "Pos", "Squad", "Gls", "Ast"
]

print(df[selected_columns].head())
```

1.5 Data Analysis and Feature Selection

```
selected_columns = ['Pos', 'Squad', 'Gls', 'Ast']

football_data = df[selected_columns]

print(football_data.head())
```

The relevant columns from the football player statistics dataset were selected for analysis. Fields such as player position, squad, goals, and assists were used to study player performance. These attributes helped create comparisons between players and clubs using different visualization techniques.

1.6 Preparing Data for Visualization

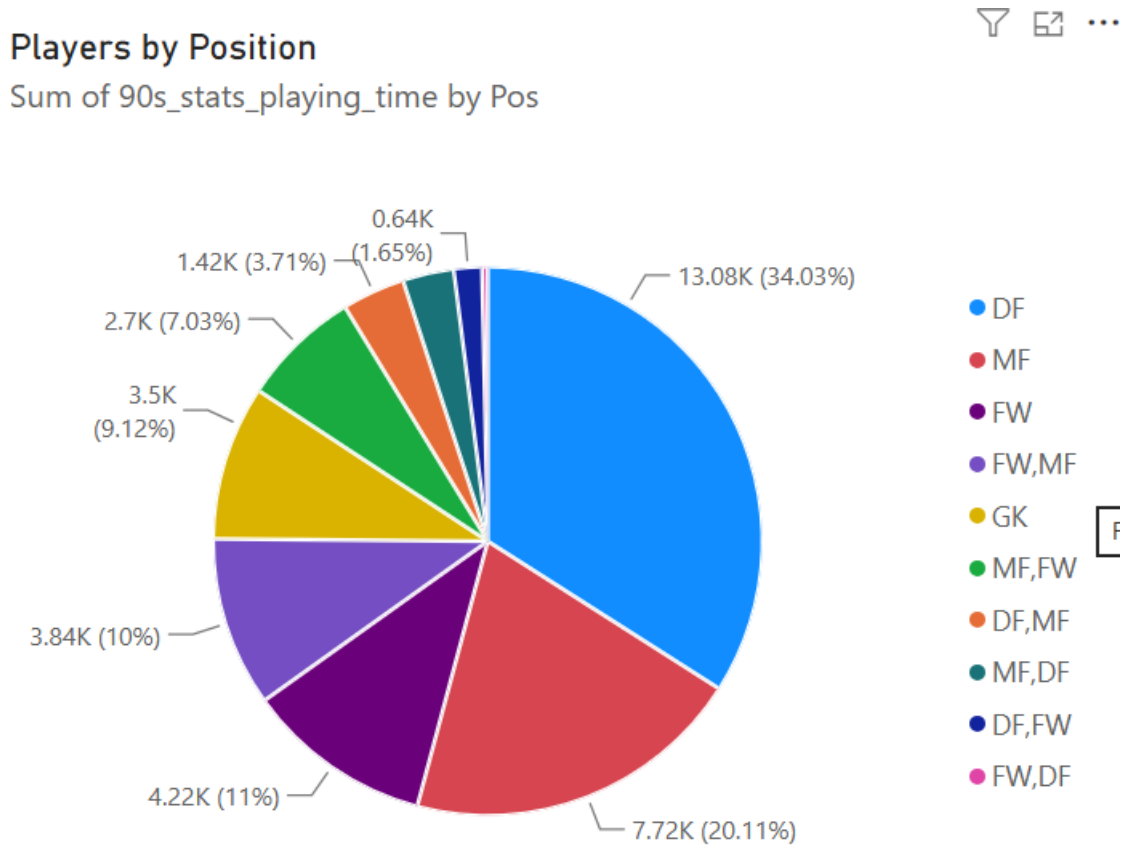
```
print(football_data.dtypes)
print(football_data.describe())
```

The selected football statistics were prepared for visualization. Numerical columns such as goals and assists were used to compare player performance, while categorical columns such as position and squad were used to group the data. The prepared dataset was suitable for creating charts and an interactive dashboard.

2. Data Visualization Using Python

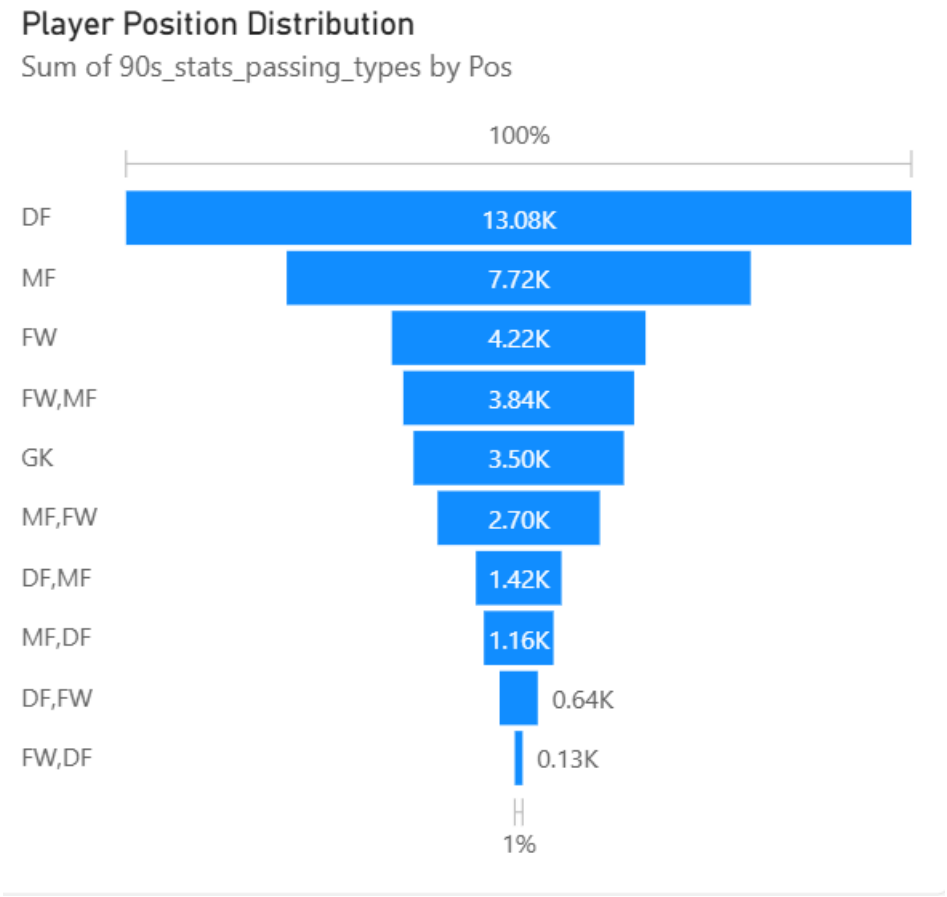
Python is a powerful programming language used for data analysis and visualization. In this project, Python libraries such as Pandas and Matplotlib can be used to process football player statistics and create visual representations of the data.

PIECHART:



Result: The pie chart represents the distribution of football players across different playing positions. It helps identify the contribution of each position category to the overall playing-time statistics. The chart provides a clear visual comparison of the different positions and makes it easier to understand their distribution in the dataset.

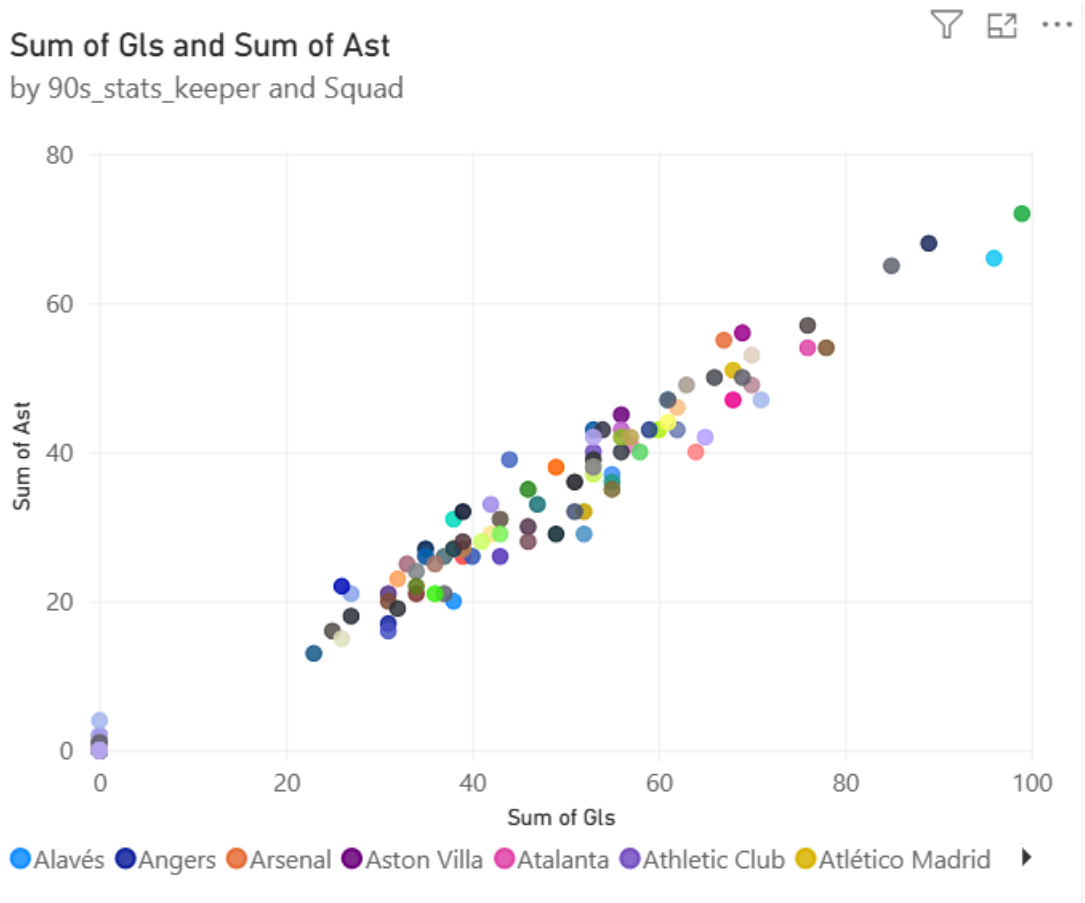
FUNNEL CHART:



Result:

The funnel chart illustrates the distribution of football players across different playing positions based on the displayed values. It helps compare the categories and identify which positions have higher and lower values. The visualization provides a clear understanding of the relative distribution of players across the different positions.

SCATTERCHART:

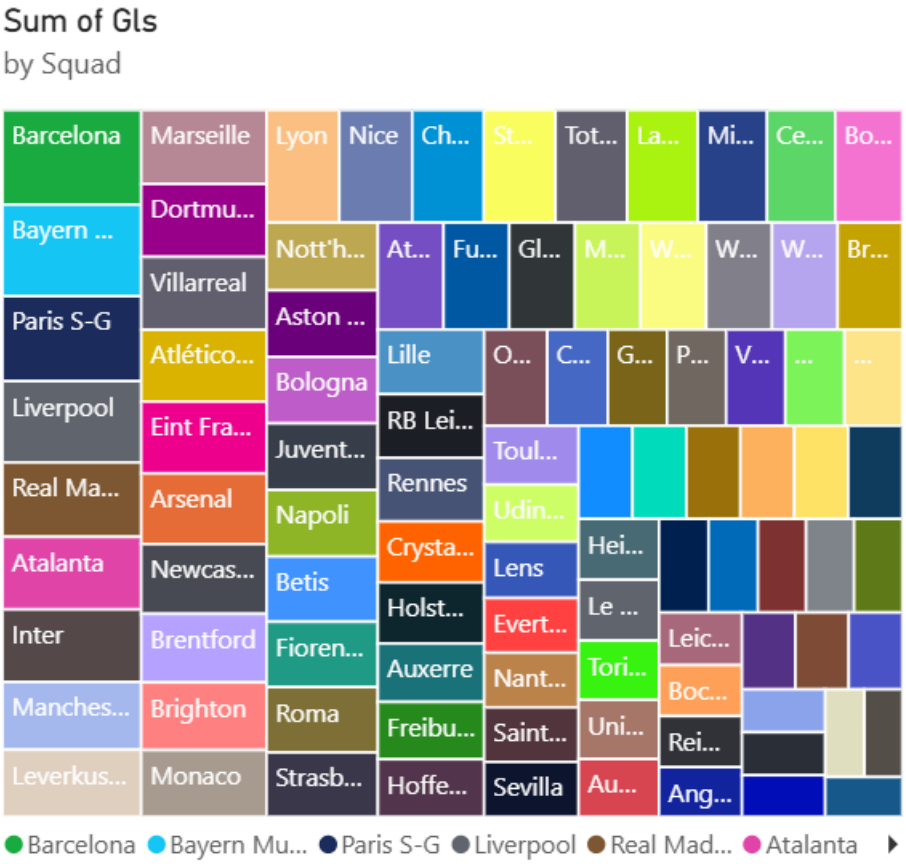


Result:

The scatter chart illustrates the relationship between the total goals and total assists recorded by different football squads. It helps compare the attacking performance of squads and identify patterns between goals and assists. This visualization makes it easier to understand the differences in team performance.

SLICER:

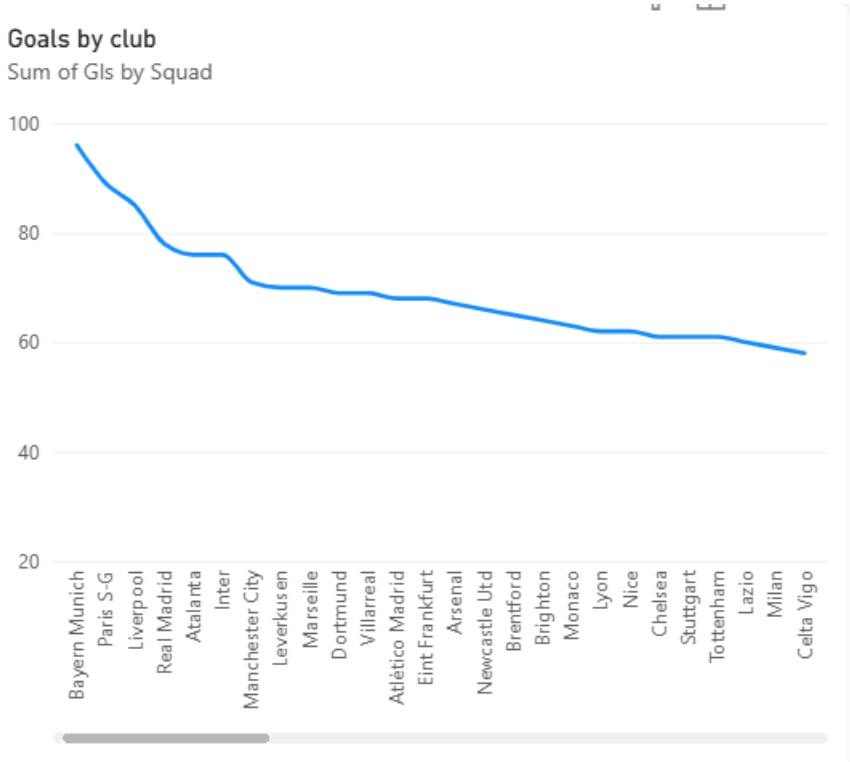
.



Result:

The slicer allows users to filter the football player dashboard based on selected criteria. It helps users view specific data and analyze player statistics according to their selection. This interactive feature makes the dashboard more user-friendly and improves data analysis.

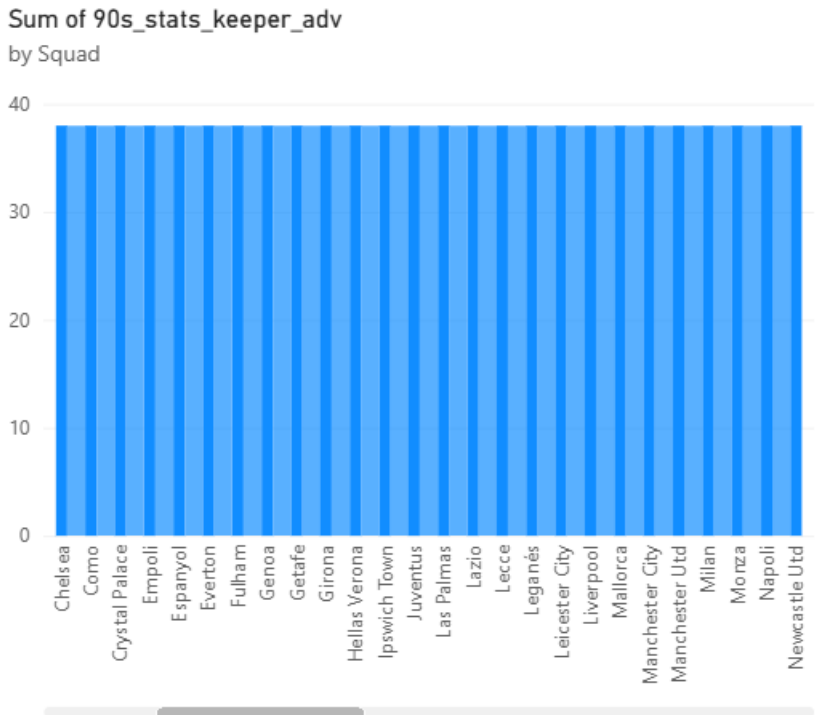
LINE CHART:



Result:

The line chart illustrates the total number of goals scored by different football clubs. It helps compare the goal-scoring performance of each club and identify differences between them. This visualization makes it easier to understand and analyze the goals scored across clubs.

TREE MAP:



Result:

The treemap chart represents the sum of goals scored by different football squads. Each rectangle represents a squad, and its size indicates the squad's contribution to the total goals. This visualization helps identify squads with higher and lower goal contributions and makes it easy to compare goal-scoring performance across squads.

3. Power BI Dashboard

Power BI was used to build a one-page interactive dashboard from the sports analysis data.

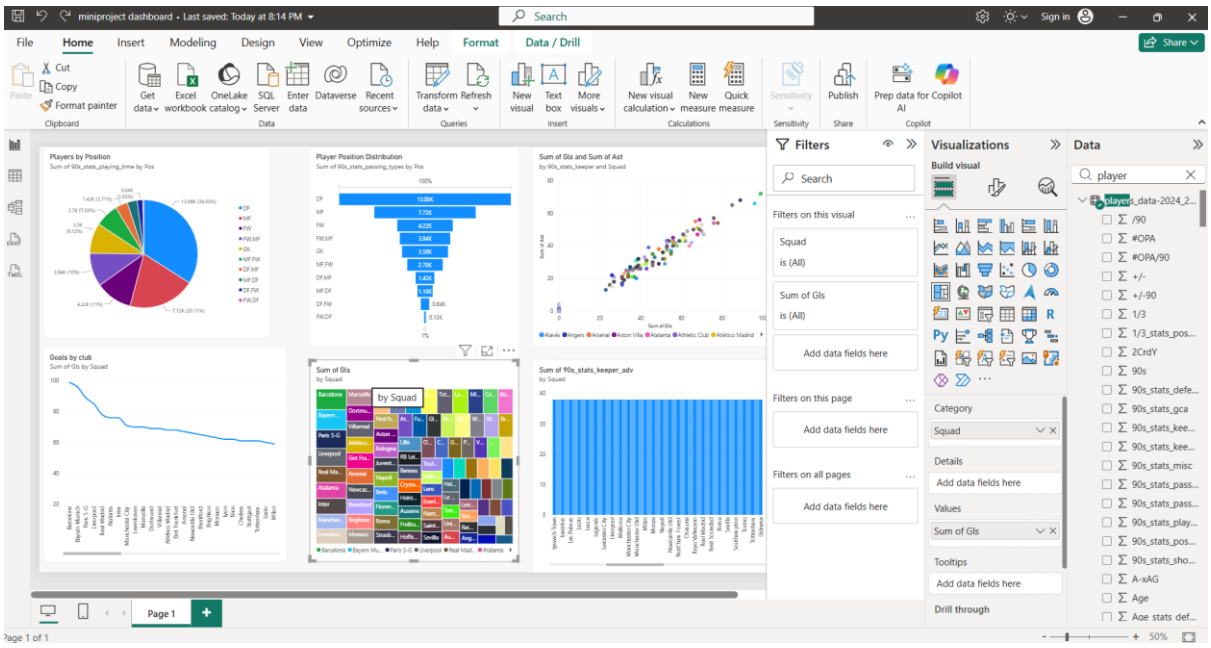


Figure 3.1: Sports Analysis Dashboard in Power BI

- **Pie Chart**: Displays the distribution of players across different playing positions.
- **Funnel Chart:** Compares the displayed values across player position categories.
- **Scatter Chart:** Shows the relationship between the sum of goals and the sum of assists by squad.
- **Line Chart:** Compares goal statistics across different clubs.
- **Treemap:** Represents the sum of goals scored by different squads using rectangles of varying sizes.
- **Bar Chart**: Compares goalkeeper-related statistics across squads.

Dashboard Insights: The visualizations help compare football statistics across positions, clubs, and squads. The scatter chart provides a visual comparison between goals and assists, while the treemap highlights differences in squad goal totals. The line chart makes club-level goal comparisons easier, and the position charts show the distribution of the displayed values across playing positions.

Result: The Power BI dashboard presents football player statistics in a clear and organized visual format. It helps users explore the dataset, compare different performance measures, and understand the available football statistics more easily.

4. Tableau Dashboard

Tableau is a data visualization tool that helps users create interactive dashboards from datasets. In this project, Tableau can be used to visualize football player performance data and compare goals, assists, player positions, club statistics, squad performance, and goalkeeper statistics.



- **Filters:** Position, Squad, and Club filters can be used to explore specific categories, if configured in the dashboard.
- **Charts**: Pie chart for players by position, funnel chart for position distribution, scatter plot for goals versus assists, line chart for goals by club, treemap for goals by squad, and bar chart for goalkeeper statistics by squad.

- **Insights:** These visualizations help compare the displayed football statistics and identify differences among positions, clubs, and squads.

- **Result:**The Tableau dashboard provides a visual overview of football player performance statistics. The charts make it easier to compare categories and understand patterns in the dataset. Interactive filters, when implemented, allow users to focus on selected positions, squads, or clubs.